

VTVL Hopper Engineering Notebook

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1 Design Requirements

1.1 Mission Objective

Design a throttleable, simple liquid bipropellant engine (and every subsystem required for it to work) for a minimal VTVL hopper. The engine exists to support a possible vehicle in the future including subsystems such as avionics, landing gear, and test instrumentation. The theoretical flight objective for the full vehicle to be designed later is controlled liftoff, hover or near-hover control authority, and safe landing.

1.2 Requirement Classification

The requirements below are limited to items that must be true before the design is considered acceptable. Quantities that set the analysis envelope (thrust class, burn time, chamber-pressure band, liftoff margin) are carried as sizing variables in Table 2 and justified in Section 1.3 **before** propellant selection. They are deliberately independent of which oxidizer or fuel wins later.

Item	Classification	Requirement / design rule
Budget	Hard	Prototype propulsion hardware target: $\leq \$5,000$ excluding propellant
Architecture	Hard	No turbopumps.
Cryogenics	Hard	No cryogenic propellant storage.
Hazard class and Legality	Hard	No propellant or working fluid that requires specialized toxicity handling, hazmat licensing, or an exotic-material compatibility program (hydrazine-family, RFNA-class, high-test-peroxide-class). Ordinary industrial hazards — oxidizers, flammable liquids, compressed gases — are accepted with standard controls.
Material access	Hard	Primary wetted structural materials shall be readily obtainable aluminum alloy or stainless steel unless a later section gives a specific compatibility basis.
Fluid connectors	Hard	Use pressure-rated commercial connector families only: AN/MS, CGA, Swagelok-compatible compression fittings, or equivalent documented hardware.
Pressure safety	Hard	Pressure-wetted custom parts shall use burst factor of safety ≥ 4 against MEOP. Purchased cylinders, valves, regulators, and fittings shall be used only within documented manufacturer ratings.
Remote operation	Hard	All oxidizer and hot-fire operations shall be controllable from a safe remote location with positive propellant isolation before personnel approach.
Throttle demonstration	Test requirement	The engine shall demonstrate repeatable operation at full thrust and at lower commanded setpoints before flight. The final throttle ratio is set after vehicle mass is known.
Combustion stability	Test requirement	No sustained hard-start, chugging, buzz, or high-frequency instability is acceptable in qualification hot-fire data. Stability criteria are finalized in the injector and test sections.
TVC interface	Layout requirement	Reserve a defined thrust-frame or gimbal interface. Cooling jacket and plumbing geometry shall not block later TVC integration.

Table 1: Design requirements for the first prototype phase.

Quantity	Status	How it is used in this notebook
Nominal thrust	Sizing variable	Analysis bracket 200–400 N for mass, cost, and static-fire planning. Final thrust is set in the engine-sizing section after dry mass and propellant mass converge.
Full-thrust liftoff margin	Design rule	Require $\frac{T_{max}}{m_0 g_0} \geq 1.2$ once m_0 is known. The numeric floor is justified from vertical control authority in Section 1.3, independent of propellant choice.
Burn duration	Design goal	30 s minimum useful demonstration; 60 s stretch target. Propellant mass follows from thrust, burn time, and $I_{sp,SL}$ after a propellant pair is selected.
Chamber pressure	Design variable	150 psia for first static-fire analysis; 200 psia carried as a performance-upgrade case if feed, cooling, and structure close. Architecture-specific feed ceilings are evaluated after the oxidizer is chosen.

Table 2: Sizing variables set before propellant selection. Brackets are mission- and architecture-class anchors only; they do not assume a winning propellant pair. Justification is in Section 1.3.

1.3 Basis for the Sizing Brackets

The brackets in Table 2 have to be fixed early enough that mass, cost, and CEA comparisons are comparable, but they must not smuggle in a propellant decision. This section sets them from mission class, pressure-fed feed physics that is common to every candidate, and published hopper scale — not from later screening results. Propellant-specific feed ceilings (self-pressurized saturation pressure, GOX regulator blowdown floor, peroxide tank ratings) are deferred until the corresponding oxidizer is under analysis.

What this section is not

These numbers are not a claim that a 17–34 kg vehicle already closes, that 150 psia is optimal for every oxidizer, or that any propellant family is preferred. They are the analysis envelope I will use for every candidate equally. If a later architecture cannot close mass or feed at these brackets, that is evidence against that architecture — not a reason to rewrite the brackets to favor it.

Two small VTVL programs publish enough numbers to use as **scale** anchors. Both happen to fly N_2O/IPA ; I record that fact for honesty and use only their thrust class, chamber-pressure regime, and liftoff loading here. Propellant selection is a separate decision later. Larger LOX/alcohol hoppers (Masten, Armadillo) are one to two orders of magnitude heavier and are recorded only in the cryogenic reconsideration (Section 2.9).

Vehicle	Scale	Published numbers	What it anchors here
Gruyère Colibri	100 kg wet, 2.45 m	Up to 1.25 kN thrust; 105 m, 60 s free flight; wet mass 100 kg [1], [2], [3].	Maximum liftoff $T/W \approx 1.27$; thrust loading ≈ 12.5 N per kg of wet mass. Scale check only — this vehicle is heavier than the class I am targeting.
AEL Snark on Gyroc 5	300 N thrust class	Throttle range 20–117% of nominal; closed-loop throttle steps at 11.0 bar(a) (≈ 160 psia) chamber pressure; measured c^* 1487 m/s and 215 s sea-level I_{sp} at O/F 5:1 [4].	Demonstrated pressure-fed chamber-pressure regime and throttle window at this thrust class; evidence that small thrusters realize 90–95% of ideal c^* .

Full-thrust liftoff margin, ≥ 1.2 . Hover is exactly $T/W = 1$, so every point above 1 is vertical control authority. At 1.2 the peak upward acceleration is about $0.2g_0 \approx 2 \text{ m/s}^2$, which arrests a 2 m/s sink rate in about one second and one meter of altitude; at 1.05 the same arrest takes about four seconds and four meters — not useful control authority on a short-hop vehicle. The margin also has to absorb the difference between ideal and delivered thrust: published small bipropellant thrusters often realize on the order of 90–95% of ideal c^*

(Snark measured 1487 m/s against an ideal near 1590 m/s at its operating point, $\approx 94\%$) [4], and any pressure-fed feed system can sag thrust at fixed valve command as tank conditions change. The upper side is bounded by throttle depth on descent: once m_0 is known, descent requires $T_{\min} < m_f g_0$, so a higher liftoff margin forces a deeper minimum throttle. With a flight propellant fraction near 25%, a 1.2 margin already implies throttling to roughly 60% of full thrust; a 2.0 margin would demand roughly 40%. That trade is propellant-agnostic at this level; how hard deep throttle is depends on the feed architecture and is evaluated later. Colibri flies near a maximum T/W of about 1.27 [1], [2], so 1.2 is a floor with a flying precedent just above it. I see no reason to buy much beyond ≈ 1.5 for a first article.

Nominal thrust, 200–400 N. With the margin rule fixed, thrust follows from the vehicle class I am willing to build. The mission is a **minimal** controlled hop under a $\leq \$5,000$ prototype-hardware budget with no turbopumps and a small-team build/test loop. That points to a wet mass in the tens of kilograms, not Colibri's 100 kg class and not a multi-kN booster. Inverting $\frac{T_{\max}}{m_0 g_0} = 1.2$ maps 200–400 N onto a wet-mass allowance of about 17–34 kg. Colibri's published loading of ≈ 12.5 N/kg maps the same thrust bracket onto ≈ 16 –32 kg, and Snark sits mid-bracket at 300 N for the same mission type [1], [4]. I am **not** using a particular oxidizer bottle or cylinder map to justify the bracket: storage mass depends on the propellant decision and will be checked after that decision. The 200 N end is the static-fire and cost floor; the 400 N end is the stretch corner for mass and cost scaling. The final flight thrust still converges in the engine-sizing section once dry mass and propellant load are known.

Chamber pressure, 150 psia baseline / 200 psia upgrade case. For a pressure-fed system, tank pressure must exceed chamber pressure plus injector drop, valve drop, and line losses. Huzel and Huang's injector-stiffness guidance of roughly 15–20% of P_c is a common preliminary allowance [5], so feed pressure is already $\approx 1.2 P_c$ before valves and lines. Published design guidance for pressure-fed engines therefore keeps chamber pressure modest so the tanks stay light: materials and systems literature commonly places pressure-fed combustion pressures in a roughly 100–200 psia band for that reason [6]. Raising P_c improves theoretical I_{sp} and shrinks the thruster for a given thrust, but it raises MEOP on every pressure-wetted part and tightens feed margin for **every** oxidizer architecture. I therefore pick 150 psia as a mid-band first analysis point inside that textbook range, and carry 200 psia as the upgrade case if feed, cooling, and structure close. Snark's demonstrated ≈ 160 psia closed-loop operation is corroborating evidence that this regime is workable at the 300 N class [4] — not a propellant lock-in. Architecture-specific ceilings (for example cold-tank saturation pressure on a self-pressurized oxidizer, or regulator delivery floor on a gaseous oxidizer) are evaluated only after that oxidizer is selected; they may force a lower P_c or a supercharge/pump decision, but they do not redefine the comparison baseline.

Burn duration, 30 s floor / 60 s stretch. This is a stated project goal, not a derived quantity: long enough to demonstrate throttle and hover-like control, short enough that propellant mass and heat soak remain first-article problems. Propellant mass is $m_p = T \frac{t}{I_{sp,SL} g_0}$ after thrust and I_{sp} are known.

I am deliberately **not** listing a numeric minimum-throttle ratio here. Descent capability is the inequality $T_{\min} < m_f g_0$ once masses are known; the ratio $\frac{T_{\min}}{T_{\max}}$ is a consequence of the liftoff margin and the propellant fraction, not an independent sizing input. The hard requirement remains the throttle demonstration test listed in Table 1.

2 Propellant

2.1 Define the Problem

Select a storable oxidizer/fuel pair for a small pressure-fed VTVL hopper. The selected pair must support repeatable throttling, safe ground testing, affordable consumables, realistic tankage, and a credible path to

injector and cooling design. The decision is not “maximize I_{sp} ”; it is “minimize total development risk while preserving enough performance for a minimal controlled hop.”

The analysis envelope is already fixed in Section 1.3 (200–400 N, $T/W \geq 1.2$, 30–60 s burns, 150 psia baseline P_c with a 200 psia upgrade case). Every candidate below is scored against that same envelope. I am not allowed to retune thrust or chamber pressure mid-trade to rescue a propellant family that fails storage, cost, or feed physics at those brackets.

Categories of analysis (these become the weighted criteria in Table 3):

- Safety and access
- Consumable and hardware cost
- Storage mass and volume
- Feed control and predictability
- Development risk
- Ignition and start risk
- Direct heritage
- Performance
- Cooling / material margin

2.2 Brainstorm Solutions

2.2.1 Hard Screens

These families fail a fixed constraint and are removed before detailed trade scoring.

Candidate	Reason for removal
LOX-based combinations	They violate the no-cryogenic first-article constraint. A LOX engine is a valid later project, not the fastest safe path to this hopper. Reconsidered quantitatively in Section 2.9; the screen stands.
Hydrazine, MMH, UDMH, NTO	The toxicity and legal burden are incompatible with the intended small-team, low-budget test program.
RFNA/alcohol	Corrosive oxidizer handling and compatibility burden are not acceptable for the preferred aluminum/stainless build.
90%+ H_2O_2 / alcohol	High-test peroxide is a specialized oxidizer with severe decomposition and contamination hazards; sourcing and handling dominate the project.

2.2.2 Feasible Solutions

Candidate family	Why it is worth considering	Initial concerns to test
N_2O /alcohols	Self-pressurizing oxidizer, compact liquid storage, non-cryogenic, active small-VTVL precedent.	Two-phase injector flow, tank-temperature sensitivity, oxidizer pressure decay during burn.
GOX/alcohols	Single-phase oxidizer metering, stable regulator-fed pressure, easier analytical injector sizing.	Large high-pressure gas volume, regulator cost, oxygen cleaning, tank mass.
H_2O_2 /alcohols	Dense storable oxidizer; 85% peroxide performance is close to N_2O /alcohol in the sea-level ambient comparison.	High-concentration peroxide sourcing, catalyst/ignition hardware, decomposition compatibility, hazardous transport.
N_2O or GOX with methanol	Methanol is cheap and historically common in rocket fuels.	Toxicity and no computed performance advantage over IPA/ethanol.
N_2O with gaseous fuels	Some spacecraft and hybrid literature exists; gaseous fuels can simplify atomization.	Poor fuel storage density and pressure-vessel mass for a 60 s ground hopper.

N ₂ O/RP-1 or kerosene	Dense fuel; common hydrocarbon fuel.	Coking/soot and regenerative cooling complexity are poor matches for a small reusable engine.
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2.3 Screen the Candidate Set

The table below is generated from the companion RocketCEA project at the Section 1.3 baseline of $P_c = 150$ psia, $P_{amb} = 14.7$ psia, and a sea-level-matched nozzle. Values are shifting-equilibrium RocketCEA/NASA CEA outputs [7]. Pressure sensitivity for survivors is checked later; it is not used to pre-select a pair.

Candidate	$I_{sp,SL}$	O/F	ϵ	T_c K	c^* m/s	Screen result
GOX / IPA	230.1 s	1.65	2.43	3237	1762	Keep as GOX finalist. Best computed sea-level impulse, but GOX storage/hardware dominates.
GOX / Ethanol	225.6 s	1.55	2.45	3187	1727	Keep as GOX finalist. Slightly lower computed I_{sp} than GOX/IPA but stronger ignition/cooling evidence.
GOX / Methanol	221.3 s	1.20	2.46	3074	1693	Remove. No advantage over GOX/ethanol or GOX/IPA, with worse toxicity.
N ₂ O / IPA	207.0 s	5.10	2.37	3120	1590	Keep. Best N ₂ O/alcohol computed result and strongest VTVL heritage.
N ₂ O / Ethanol	204.8 s	4.65	2.38	3064	1573	Keep. Strong static-fire literature and better coolant conductivity.
85% H ₂ O ₂ / IPA	203.2 s	5.65	2.37	2553	1562	Near miss only. Dense oxidizer but sourcing/catalyst/decomposition burden is high.
N ₂ O / Methanol	203.0 s	3.50	2.38	2972	1559	Remove. No performance advantage and worse toxicity.
85% H ₂ O ₂ / Ethanol	201.1 s	5.00	2.36	2507	1547	Near miss only; same peroxide concerns.
70% H ₂ O ₂ / IPA	186.9 s	7.20	2.31	2184	1441	Remove. Low performance without solving peroxide access/catalyst risk.
70% H ₂ O ₂ / Ethanol	185.0 s	6.30	2.30	2139	1426	Remove. Low performance without solving peroxide access/catalyst risk.

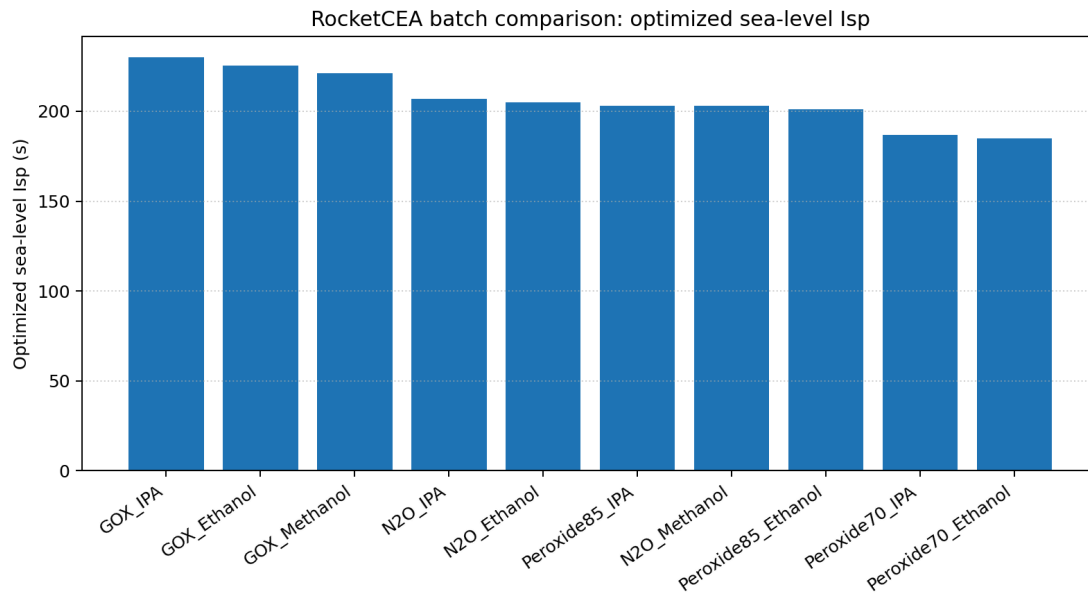


Figure 1: Generated RocketCEA comparison for optimized sea-level impulse at $P_c = 150$ psia, $P_{amb} = 14.7$ psia, shifting-equilibrium chemistry, and $P_e \approx P_{amb}$ nozzle sizing [7].

2.4 Quantitative Sourcing and System Sizing

Pricing below is a planning model, not a purchase instruction. Industrial gas prices vary by region, account status, deposit/rental policy, hazmat shipping, and cylinder exchange rules. The useful engineering comparison is the order of magnitude: GOX consumables can be cheap, but the storage/regulator system is expensive and heavy; N_2O consumables are more expensive per kilogram, but the liquid storage is much smaller and the tank is also the pressure source. I should replace these bands with local quotes before procurement, but the bands below are anchored to published prices — including the Friends of Amateur Rocketry propellant list, which is the closest thing to a posted market price for amateur-scale rocket oxidizers and includes tax and delivery to a real test site [8].

Material	Practical acquisition route	Planning price	Design note
N_2O	Racing-gas cylinder from an amateur test site or gas supplier; motor-sport refill	\$12–\$22/kg	FAR sells a 56 lb racing-grade cylinder with dip tube for \$292, about \$11.5/kg [8]; motorsport refills run \$8.49–\$9.79/lb (\$18.7–\$21.6/kg) [9]. Higher one-off retail fills are reported.
GOX	Welding oxygen cylinder exchange or industrial gas supplier	\$4–\$16/kg	Welding-trade guides put small-cylinder refills/exchanges around \$4–\$12/kg [10]; FAR sells a filled 125 cf cylinder for \$71.84 including tax and delivery, about \$15/kg [8].
LOX	Industrial gas supplier dewar; also sold at amateur test sites	\$2–\$4/kg	Screened out for the first article but priced for honesty: FAR sells 230 L (roughly 262 kg) for \$540 [8], and a trade source reports a \$200 quote for a 180 L dewar [11]. Dewar deposit/rental and boiloff excluded. See Section 2.9.
99% IPA	Hardware, janitorial, lab, or chemical supplier	\$8–\$12/kg	Representative 99% IPA is \$23.50/gal before shipping/hazmat; 70% rubbing alcohol is not acceptable [12].
Anhydrous ethanol	Industrial/lab chemical supplier; denatured 200-proof acceptable if compatible	\$13–\$30/kg	Denatured 200-proof ethanol can be near \$40–\$57/gal, while tax-paid absolute ethanol can be much higher [13], [14].

70–85% H ₂ O ₂	Specialty chemical supplier; high-concentration grades usually quote/hazmat	\$20–\$60+/kg placeholder	Public lower-concentration lab peroxide pricing is visible, but 70% is specialty/quote and not ordinary pool-supply grade [15], [16], [17].
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Case	Thrust	Total prop.	Oxidizer	Fuel	Ox storage	Propellant cost
N ₂ O / IPA	200 N	5.91 kg	4.94 kg	0.97 kg	7.9 L	\$67–\$120
N ₂ O / IPA	400 N	11.82 kg	9.88 kg	1.94 kg	15.8 L	\$134–\$241
N ₂ O / Ethanol	200 N	5.97 kg	4.92 kg	1.06 kg	7.9 L	\$73–\$140
N ₂ O / Ethanol	400 N	11.95 kg	9.83 kg	2.11 kg	15.7 L	\$145–\$280
GOX / IPA	200 N	5.32 kg	3.31 kg	2.01 kg	17.5 L	\$29–\$77
GOX / IPA	400 N	10.64 kg	6.62 kg	4.01 kg	34.9 L	\$59–\$154
GOX / Ethanol	200 N	5.42 kg	3.30 kg	2.13 kg	17.4 L	\$41–\$117
GOX / Ethanol	400 N	10.85 kg	6.59 kg	4.25 kg	34.8 L	\$82–\$233

The GOX volume in the table assumes storage at 2200 psia and 293 K using the ideal-gas law plus a 5% design margin; the vessel map below uses the 2265 psia DOT service rating of the actual cylinders, and that 65 psi difference sits inside the margin. The N₂O volume assumes saturated-liquid storage near room temperature with a 20% ullage/design margin. The 750 kg/m³ N₂O density is deliberately a warm-tank planning value: saturated-liquid density is roughly 745–790 kg/m³ over 20–25°C and falls steeply toward the critical point, so a hot Las Vegas tank holds less than a cool one [18]. These are comparison numbers, not final pressure-vessel dimensions; candidate pressure vessels still need manufacturer ratings, valve details, fill rules, and compatibility checks [19].

The 400 N, 60 s case is the stretch corner of the requirement, not the requirement, so it must not size the architecture by itself; the honest way is to map every corner of the thrust/burn-time envelope onto real, priced vessels. Stored GOX assumes blowdown from 2265 psia service pressure to a 500 psia regulator floor, leaving about 78% of the fill usable; the N₂O planning load is 1.25× the burned mass to cover retained vapor, pressure sag near depletion, and fill tolerance [19].

Case	GOX burned / stored	GOX vessel (price, tare)	N ₂ O burned / loaded	N ₂ O bottle (price, tare)
200 N, 30 s	1.65 / 2.12 kg	80 cf steel (\$143.50)	2.47 / 3.09 kg	10 lb bottle (\$314.99, 6.3 kg)
200 N, 60 s	3.30 / 4.23 kg	125 cf steel (\$194.50, 26.3 kg)	4.94 / 6.18 kg	15 lb bottle (\$374.99, 8.1 kg)
400 N, 30 s	3.30 / 4.23 kg	125 cf steel (\$194.50, 26.3 kg)	4.94 / 6.18 kg	15 lb bottle (\$374.99, 8.1 kg)
400 N, 60 s	6.59 / 8.46 kg	250 cf steel (\$274.80, 49.0 kg)	9.88 / 12.36 kg	2 × 15 lb bottles (\$749.98, 16.1 kg)

Cylinder and bottle picks, prices, and tares are current public listings [20], [21], [22], [23]. Read this table two ways. As ground-support hardware, GOX storage is cheap at every requirement point: the 30 s floor case is one \$143.50 welding cylinder, and only the full stretch case forces the 250-class cylinder. As onboard hardware, the same column is disqualifying: the mapped steel cylinders carry roughly 26–49 kg of steel to deliver 3.3–6.6 kg of usable oxygen, about 7 kg of vessel per kilogram of oxidizer against about 1.5 kg/kg for the nitrous bottles. Aluminum and composite oxygen cylinders improve the ratio but remain far from the bottle numbers and cost more, and breathing-air composite cylinders (SCBA, paintball) are not rated for oxygen service and are not a shortcut.

All of the propellant loads above use ideal CEA I_{sp} , and no real thruster delivers it: Snark measured about 94% of ideal c^* at its operating point [4]. Carrying a planning 92% c^* efficiency and 97% C_F efficiency – about 89% of ideal I_{sp} – raises every load by about 12%:

Case	Ideal $I_{sp,SL}$	Delivered (plan)	200 N, 60 s load	400 N, 60 s load
N ₂ O / IPA	207.0 s	184.7 s	5.91 → 6.62 kg	11.82 → 13.25 kg
N ₂ O / Ethanol	204.8 s	182.8 s	5.97 → 6.69 kg	11.95 → 13.39 kg
GOX / IPA	230.1 s	205.3 s	5.32 → 5.96 kg	10.64 → 11.92 kg

GOX / Ethanol	225.6 s	201.3 s	5.42 → 6.08 kg	10.85 → 12.16 kg
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The 30 s cases keep their mapped vessels with margin. Every 60 s corner lands 1–2% over its vessel instead: the delivered-basis N_2O loads become 6.92 kg against the 15 lb bottle's 6.80 kg capacity and 13.85 kg against 13.61 kg for the bottle pair, and the GOX 60 s fills run about 0.6% over the usable blowdown fraction. That overshoot is smaller than the $1.25\times \text{N}_2\text{O}$ loading allowance it overlaps — the allowance already double-books part of the same shortfall — so the vessel map stands, but the ten-fire calibration budget scales with the same 12%, to roughly \$750–\$1,350.

The vessel is not where the architectures diverge on cost, so the next table itemizes the full oxidizer-side feed system at the 200 N, 60 s reference. Allowance rows are marked as such; everything else is a current public listing.

Item	GOX / alcohol	N_2O / alcohol
Storage vessel	125 cf steel CGA-540 cylinder: \$194.50, 26.3 kg tare [20].	15 lb aluminum bottle with high-flow valve: \$374.99, 8.1 kg tare [23].
Pressure regulation	High-flow high-delivery oxygen regulator. Harris 3000-2500 lists \$699 [24]; Victor SR4J-540 lists \$1,077, rated 16,600 scfh against roughly 5,300 scfh needed at 200 N and 10,500 scfh at 400 N [25].	None. Tank vapor pressure is the feed source.
Run valve and plumbing	Oxygen-clean actuated run valve, check valve, relief, CGA-540 pigtail: \$250–\$450 allowance (quote items).	Motorsport N_2O solenoid or actuated ball valve plus -AN plumbing: \$250–\$550 allowance (quote items).
Fill and ground handling	Cylinder exchange at the supplier; no owned fill hardware.	Fill adapter and scale: \$100–\$250 allowance, or per-fill service at a speed shop [9].
Cleanliness	Every wetted part oxygen-cleaned per CGA G-4.1: \$50–\$150 consumables and verification [26].	No separate line item, but the same discipline applies: hydrocarbon contamination catalyzes N_2O decomposition [27].
Subtotal	\$1,190–\$1,870	\$725–\$1,175

Shared subsystems are excluded from both columns because both architectures need them: fuel tank, fuel-side pressurant, commodity pressure transducers (\$30–\$80 class), harness, and test stand. One common claim needs correcting here: N_2O does not eliminate the pressurant bottle, only the oxidizer-side one. The AEL Snark vehicle fed IPA from a nitrogen-pre-pressurized tank, and this project will need the same small N_2 system either way [4].

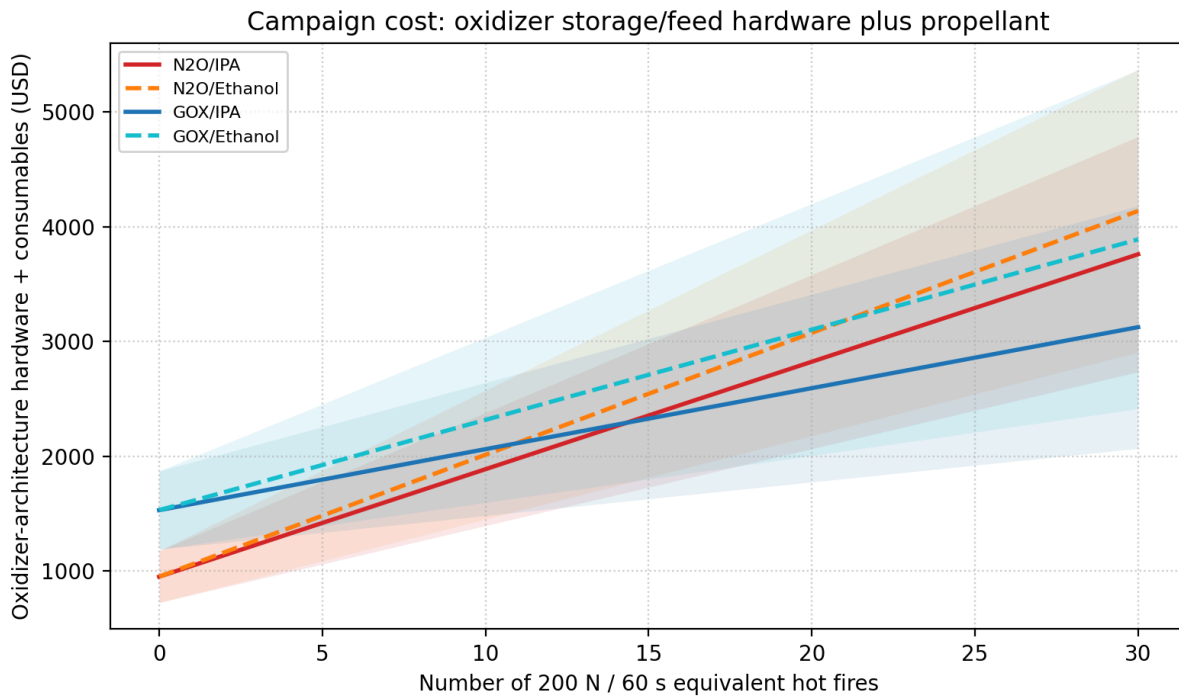


Figure 2: Oxidizer-architecture hardware plus consumables against number of 200 N, 60 s hot fires; mid-band lines with low/high shading [19].

The hardware premium and the consumables premium pull in opposite directions, so the honest comparison is the whole campaign. The GOX architecture costs roughly \$470–\$700 more up front, almost all of it regulator and oxygen-clean components rather than the cylinder. The N₂O architecture pays roughly \$38–\$43 more per test than GOX/IPA and \$4–\$26 more than GOX/ethanol, whose expensive fuel erodes most of the cheap-gas advantage. On mid-band assumptions the GOX premium pays back after about 15 hot fires against GOX/IPA and only after about 40 against GOX/ethanol. For a pure static-fire program, GOX/IPA is the cheapest campaign. What that framing misses is the flight article: the compact bottle in the mapping table above is the only storage option in this set that a hopper can plausibly lift, and it doubles as the pressurization system.

2.5 Peroxide and Methanol Near-Misses

85% H₂O₂/alcohol is not dismissed because of sea-level performance alone. In the baseline comparison it is close to N₂O/alcohol: 85% H₂O₂/IPA gives 203.2 s, only 3.8 s below N₂O/IPA [7]. It is removed because the supporting architecture is worse for this project: high-concentration peroxide sourcing is specialized, material compatibility is narrow, decomposition contamination can be violent, and practical engines often need a catalyst or dedicated ignition/decomposition hardware [17], [28].

70% H₂O₂ is not ordinary pool-supply grade in the sense needed here. Even if a 70% source is found, sea-level ambient-corrected performance is about 185–187 s at the baseline condition, while retaining the peroxide compatibility and handling burden [7].

Methanol combinations are also removed. Methanol does not outperform the matching IPA/ethanol cases in the baseline comparison, and the CDC/NIOSH pocket guide lists skin absorption as an exposure route, OSHA PEL of 200 ppm, and symptoms including visual disturbance and optic nerve damage [29]. That is a poor trade for no performance advantage.

2.6 Analyze Survivors

2.6.1 N₂O / IPA

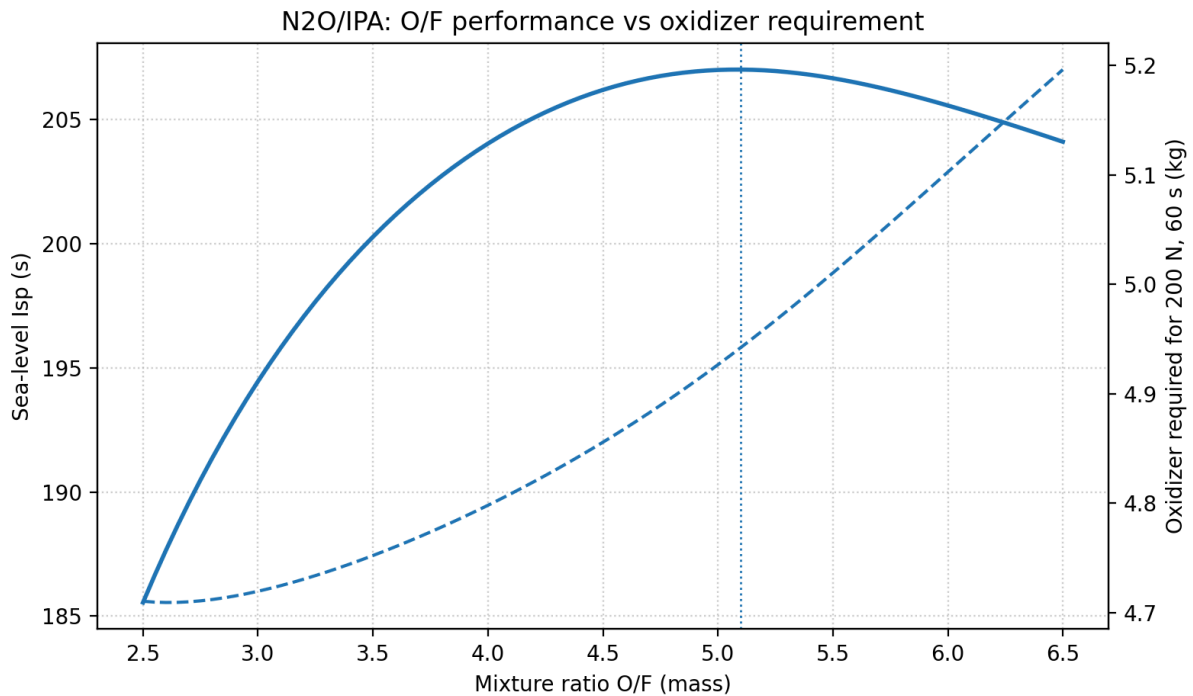


Figure 3: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

The O/F curve also shows why “run fuel-rich to save oxidizer” is not a large win for this pair. At the 200 N, 60 s reference burn, moving from O/F = 5.10 to O/F = 4.50 cuts N₂O from 4.94 kg to 4.86 kg, but loses 0.8 s of computed sea-level I_{sp} and adds about 0.11 kg of IPA. At O/F = 4.00, the oxidizer savings are only about 0.14 kg while the penalty grows to about 3.0 s. I would not move far below the optimum only for oxidizer mass; I would shift O/F only if injector stability, cooling, or hot-fire data asks for it [7].

N₂O/IPA survives screening as a strong contender for the hopper mission. The sea-level ambient-corrected computed performance is $I_{sp,SL} = 207.0$ s at O/F = 5.10 and $\varepsilon = 2.37$ [7]. That is not high compared with vacuum-optimized numbers, but it is enough for a minimal hop and avoids the hardware burden of high-pressure GOX storage.

The strongest external evidence for this pair is direct VTVL heritage. Gruyère Space Program’s Colibri vehicle uses an N₂O/IPA bipropellant engine and is documented as a 2.45 m VTVL demonstrator with up to 1.25 kN thrust [2]. European Spaceflight reported the October 18, 2024 free flight to 105 m altitude, 30 m lateral translation, and 60 s duration [3]. That is the closest public analog to this project. Formal selection waits until the architecture, risk, and cost comparisons below are complete.

Decision Standing after analysis. Compact self-pressurizing oxidizer storage, no oxidizer regulator, cheap fuel, and direct small-VTVL precedent. The open risk is injector/feed calibration for flashing N₂O (Section 2.6.3), not sea-level I_{sp} .

2.6.2 N₂O / Ethanol

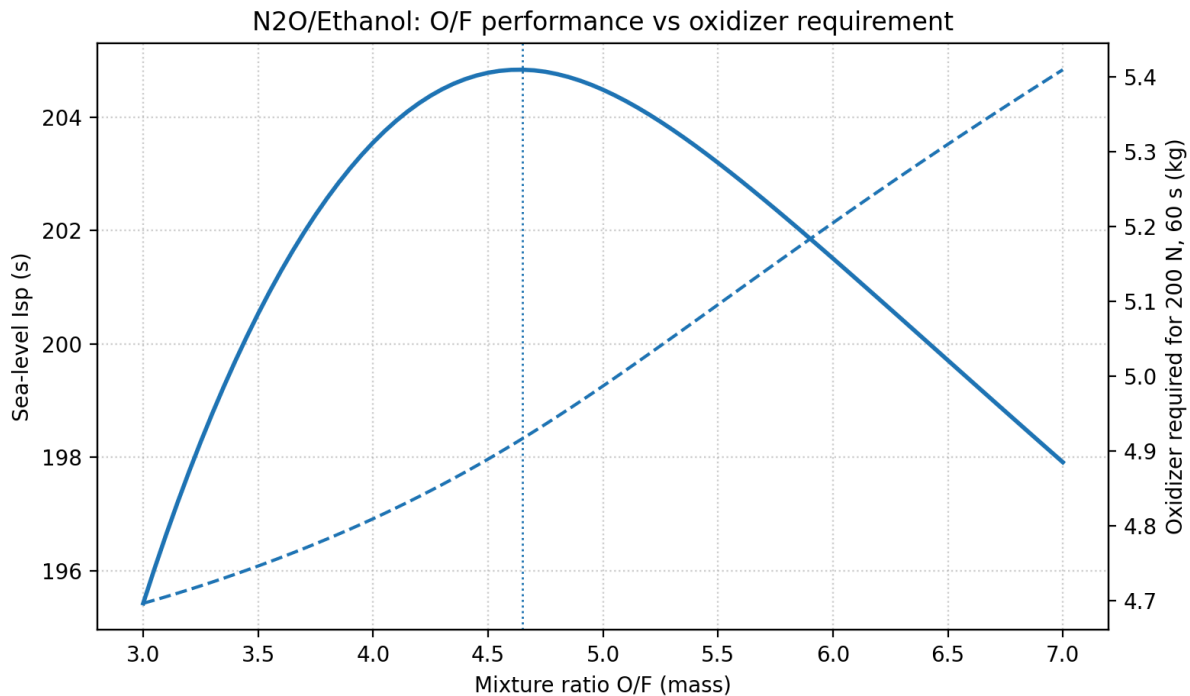


Figure 4: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

For N₂O/ethanol the curve is broad near the optimum. O/F = 4.20 reduces N₂O from 4.92 kg to 4.84 kg and costs only 0.6 s, so a small fuel-rich bias is acceptable if it helps cooling or calibration. Going to O/F = 4.00 saves only about 0.11 kg oxidizer and already loses 1.3 s; O/F = 3.50 saves about 0.17 kg but loses 4.3 s. The mass benefit is too small to justify a large performance departure by itself [7].

N₂O/ethanol is a close peer to N₂O/IPA: broader academic static-fire literature and better coolant thermal conductivity, at nearly the same computed impulse. The sea-level ambient-corrected computed performance is $I_{sp,SL} = 204.8$ s at O/F = 4.65 and $\varepsilon = 2.38$ [7]. The difference from N₂O/IPA is only 2.2 s at the sizing-basis baseline, so performance alone should not decide between these two fuels.

Ethanol has slightly lower specific heat but higher thermal conductivity than IPA at room temperature; the coolant trade is therefore not one-sided. IPA has slightly better bulk heat capacity, while ethanol has better film-side conduction margin [30].

Decision Standing after analysis. Natural fuel alternate on the same N₂O architecture. Prefer ethanol if the design is intentionally aligned with N₂O/ethanol literature or if regen-cooling margin dominates fuel procurement convenience.

2.6.3 N₂O Architecture Risk: Feed, Injector, and Environment

This is the most important risk record in the propellant decision, because it describes the one problem the N₂O finalists cannot close on paper. N₂O is stored as a saturated liquid at its own vapor pressure. Resonac lists high-purity N₂O vapor pressure as 5.24 MPa at 20°C (≈ 760 psia) and gives a critical temperature of 36.41°C with critical pressure 7.24 MPa [31]. This enables self-pressurization, but it also means feed pressure changes strongly with tank temperature: saturation pressure is already about 590 psia at 10°C and 457 psia at 0°C [31]. It also changes during the burn: as liquid leaves, the remaining propellant evaporates to refill the ullage, the

tank self-cools, and feed pressure sags, so thrust decays at a fixed valve position [32]. A 60 s hover is exactly the burn where that matters.

That temperature dependence is the N_2O -specific chamber-pressure ceiling that Section 1.3 deferred. Throttle-valve drop, injector drop, and chamber pressure must all fit under the **coldest credible end-of-burn tank pressure**, not under the 760 psia of a 20°C full tank. Holding $\approx 20\%$ of P_c for injector stiffness [5], a 150 psia chamber asks for ≈ 180 psia at the injector inlet and still leaves roughly 270 psi of valve authority on a 0°C morning before self-cooling sag; a 300 psia chamber asks for ≈ 360 psia and leaves under 100 psi on the same morning. The 150 psia baseline therefore fits self-pressurized N_2O with usable throttle margin; anything much above the 200 psia upgrade case needs a warm-tank-only envelope or oxidizer supercharge.

The sag itself can be bounded with an equilibrium energy balance. Draining the 200 N, 60 s load of 4.94 kg opens about 6.3 L of ullage, and refilling it with saturated vapor near 160 kg/m³ evaporates roughly 1.0 kg of liquid at ≈ 170 kJ/kg of latent heat [18]. If only the remaining liquid supplies that heat (≈ 3.7 kg average at $c_p \approx 3$ kJ/kg·K), the tank cools about 15 K over the burn; letting an 8 kg aluminum bottle share the load softens that to about 9 K. From a 20°C start the end-of-burn saturation pressure lands near 520–600 psia — a sag of roughly 160–240 psi, about 3–4 psi/s, still ≈ 340 psi above the 180 psia injector-inlet ask. From a 0°C start the end state is ≈ 370 –410 psia and the remaining valve authority shrinks to ≈ 190 –230 psi. These are equilibrium bounds and real tanks lag them [32], so the number to trust is the cold-flow measurement — but the bound already says the 150 psia baseline survives the coldest credible morning with authority to spare.

► Las Vegas design-basis implication

A N_2O tank in direct summer sun can approach or exceed the 36.4°C critical temperature. The baseline control map should be built around measured tank temperature and pressure, and the pressure-vessel MEOP check should use the highest credible tank temperature, not just nominal room temperature. Testing in shade or morning conditions is not a convenience; it is part of keeping the feed model inside the calibrated regime.

When saturated liquid N_2O accelerates through valves and injector passages, pressure reduction can initiate flashing before the chamber. The standard single-phase incompressible equation is therefore only a bounding model:

$$\dot{m} = C_d A \sqrt{2\rho\Delta P}$$

N_2O injector design should bracket the flow with SPI and HEM and then use a non-equilibrium model such as Dyer/NHNE for the design estimate. Published work on self-pressurizing propellant flow and N_2O injectors supports this SPI/HEM/NHNE framing [33], [34], [35], [36].

Model	Assumption	Expected bias	Use in this project
SPI	Single-phase incompressible liquid.	Often overpredicts flashing N_2O flow.	Upper-bound geometry check only.
HEM	Two-phase thermodynamic equilibrium.	Often lower-bound for short injector residence times.	Lower-bound flow check.
Dyer / NHNE	Non-homogeneous, non-equilibrium interpolation between SPI and HEM using residence/bubble-growth times.	Best practical pre-test estimate for short rocket-injector passages.	Design-reference model, then calibrate by hot fire.
Empirical map	Valve command, tank state, and chamber response fitted from tests.	Most reliable inside tested range.	Required for throttle control.

The AEL Snark work is the most relevant single data point because it is a 300 N N_2O /IPA thruster built for a VTVL vehicle: this thrust class, this propellant pair, this mission [4]. Two of its measured results show how

far the system sits from textbook hydraulics. First, the N_2O injector pressure drop was found to be almost independent of massflow, because flash-boiling in the control valve and flash-boiling in the injector elements balance each other; the familiar square-law relation between orifice pressure drop and massflow simply does not hold. Second, chamber pressure varied almost linearly with the throttle-valve command, and the closed-loop controller was designed around a second-order transfer function identified from test data, not derived from first principles [4]. At O/F near 5 the oxidizer is more than 80% of the total flow, so the side of the engine that resists analysis is also the side that dominates it.

The practical consequence is concrete. I cannot pre-compute a valve schedule, an injector pressure-drop margin, or a throttle map for the N_2O engine and trust them. I can only bracket them with SPI/HEM/NHNE and then buy the real answer with a calibration matrix of hot fires across tank temperatures and throttle setpoints. That campaign is a real budget line — about \$670–\$1,200 of propellant for ten 200 N, 60 s fires at the sourced prices — and it is the main technical reason not to over-score N_2O just because its tanks are compact.

N_2O also carries a hazard GOX does not: it is a monopropellant-capable oxidizer whose ullage can decompose energetically if an ignition source reaches it, and hydrocarbon contamination catalyzes that decomposition [27]. The mitigation overlaps with GOX practice — oxidizer-clean plumbing, no oil or grease, care with adiabatic compression — so choosing N_2O does not buy exemption from cleanliness discipline. The fair counterpoint from the same work is that N_2O decomposition kinetics are roughly six orders of magnitude slower than H_2O_2 at comparable conditions, which is part of why it remains the standard amateur oxidizer [27].

Ignition is the start-transient face of the same hazard. N_2O releases its oxygen only after dissociation, so the pair does not light from a bare spark the way GOX/alcohol can; a dedicated igniter must be proven lit before the main oxidizer valve opens. A slow start lets unburned propellant accumulate, and a hard start feeding a monopropellant-capable ullage is exactly the energetic event class documented for N_2O systems [27]. GOX/ethanol, by contrast, has a published ignition characterization near this chamber pressure [37]. That difference is scored as its own criterion in the selection matrix below, and the igniter itself is scoped in Section 4.

2.6.4 GOX / Ethanol

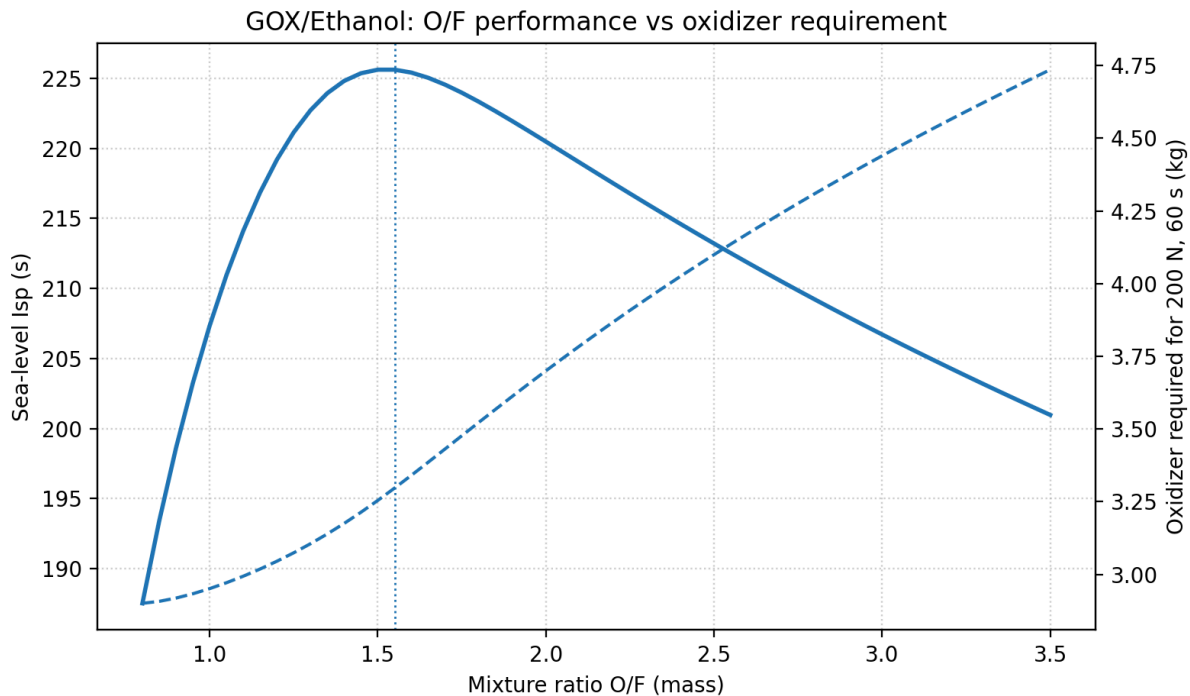


Figure 5: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

GOX/ethanol has a more useful fuel-rich knob because oxygen storage is the GOX penalty. At the 200 N, 60 s reference burn, $O/F = 1.30$ saves about 0.19 kg GOX, or 5.8%, with a 2.9 s computed I_{sp} loss; this may be worth checking if tank volume is the limiting item. Pushing to $O/F = 1.20$ saves 0.25 kg GOX, or 7.6%, but loses 6.4 s and increases ethanol mass. The useful trade window is narrow, around $O/F = 1.30$ – 1.55 , not simply “as fuel-rich as possible” [7].

GOX/ethanol survives as the cleanest single-phase oxidizer alternative if the N_2O two-phase injector problem becomes unacceptable. The sea-level ambient-corrected computed performance is $I_{sp,SL} = 225.6$ s at $O/F = 1.55$ and $\epsilon = 2.45$ [7]. It gives simpler single-phase oxidizer metering and stable regulator-fed pressure; the system-level cost of that simplicity is quantified in Section 2.4.

A NASA ignition characterization program tested GOX/ethanol at 150 psia and $O/F = 1.8$, making this the best-documented GOX/alcohol reference point near the sizing-basis chamber pressure [37]. Oxygen-side hardware must be oxygen-cleaned and kept free of hydrocarbon contamination; the oxygen-cleaning and material-control burden is real, not paperwork [26].

Decision Standing after analysis. Preferable GOX fuel on nearby ignition-validation evidence. System-level storage, regulator mass, and oxygen-clean hardware remain the architecture penalties, independent of the alcohol choice.

2.6.5 GOX / IPA

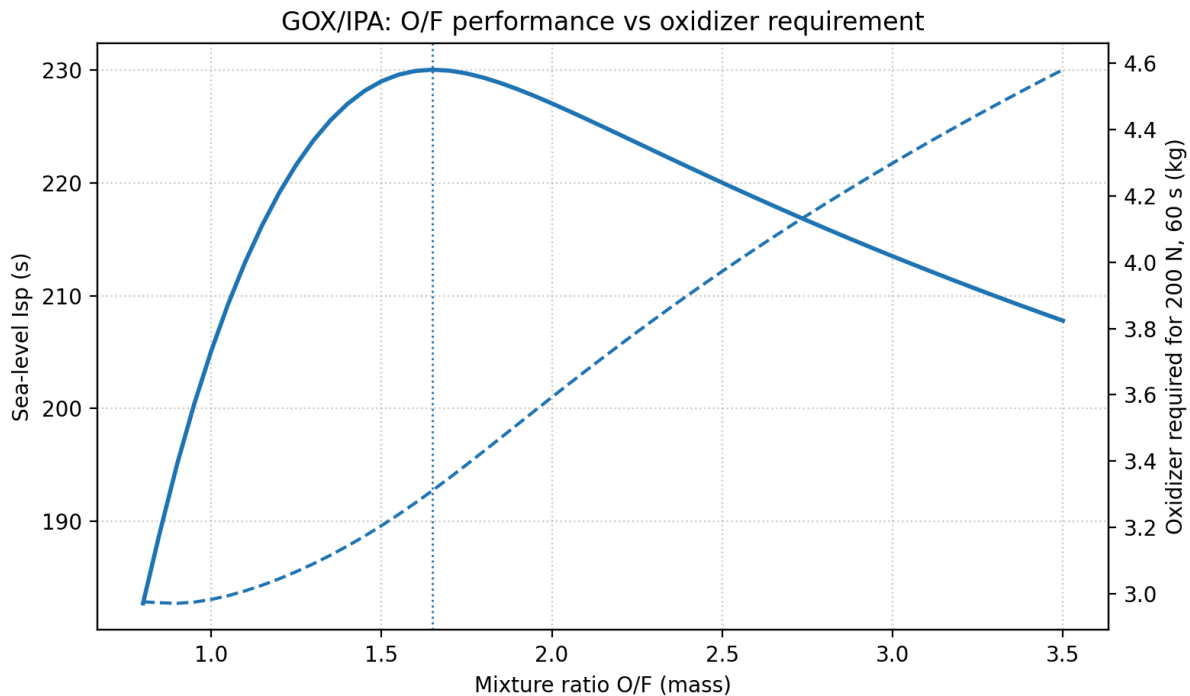


Figure 6: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

GOX/IPA is less forgiving on the rich side. Dropping O/F from 1.65 to 1.40 saves about 0.17 kg GOX, or 5.1%, while losing 3.0 s, which could be tolerable if cylinder volume is marginal. Dropping to O/F = 1.20 saves only another 0.10 kg GOX beyond that but loses 10.9 s total, so the storage saving is not enough to justify the performance loss. If GOX/IPA is used, I would stay near O/F = 1.40–1.65 unless hot-fire stability points elsewhere [7].

GOX/IPA has the best sea-level ambient-corrected computed performance in this set: $I_{sp,SL} = 230.1$ s at O/F = 1.65 and $\varepsilon = 2.43$ [7]. The fuel is cheaper than anhydrous ethanol, and per Section 2.4 that makes GOX/IPA the cheapest campaign in the whole set on consumables. The expensive pieces are the high-flow oxygen regulator, oxygen-compatible valves, and cleaning, not the cylinder; the itemized architecture cost is in Section 2.4. The main GOX advantage is single-phase oxygen metering, which neither alcohol changes; GOX/ethanol has stronger nearby ignition-validation evidence.

Decision Standing after analysis. Highest computed I_{sp} and cheapest pure static-fire consumables path. Does not solve the GOX storage/regulator flight penalty.

2.7 Pump Option Check

The hard rule is no turbopumps, not no pumps. I am still keeping the first-article survivor comparison pressure-fed because the pump does not remove the dominant oxidizer problems. For N_2O , self-pressurization is the benefit, and published injector work still treats pump/supercharge conditions as ways to set tank state or avoid cavitation, not as a way to eliminate flashing physics [34]. For GOX, a small electric liquid pump is not available because cryogenic LOX is screened out, and a gas compressor/feed pump would add oxygen-clean rotating hardware, battery power, controls, and heat without solving the large gas-storage requirement.

A small electric fuel pump remains allowed. It may become attractive later if the fuel tank or fuel pressurant mass dominates the vehicle, especially because IPA/ethanol are ordinary liquids compared with the oxidizers. I am not adding it to the baseline yet because it adds a controller, battery/current margin, bypass or relief path, transient calibration, and a new single-point failure while leaving the oxidizer architecture unchanged.

The main performance lever a pump or supercharge would buy is chamber pressure beyond the pressure-fed feed-margin band in Section 1.3. Figure 7 prices that lever for the four alcohol survivors after screening: about +11–12 s of sea-level I_{sp} from 150 to 200 psia and +24–28 s from 150 to 300 psia, with diminishing returns. That gain is worth carrying as the 200 psia upgrade case; it is not worth new oxidizer-side rotating hardware for the first article.

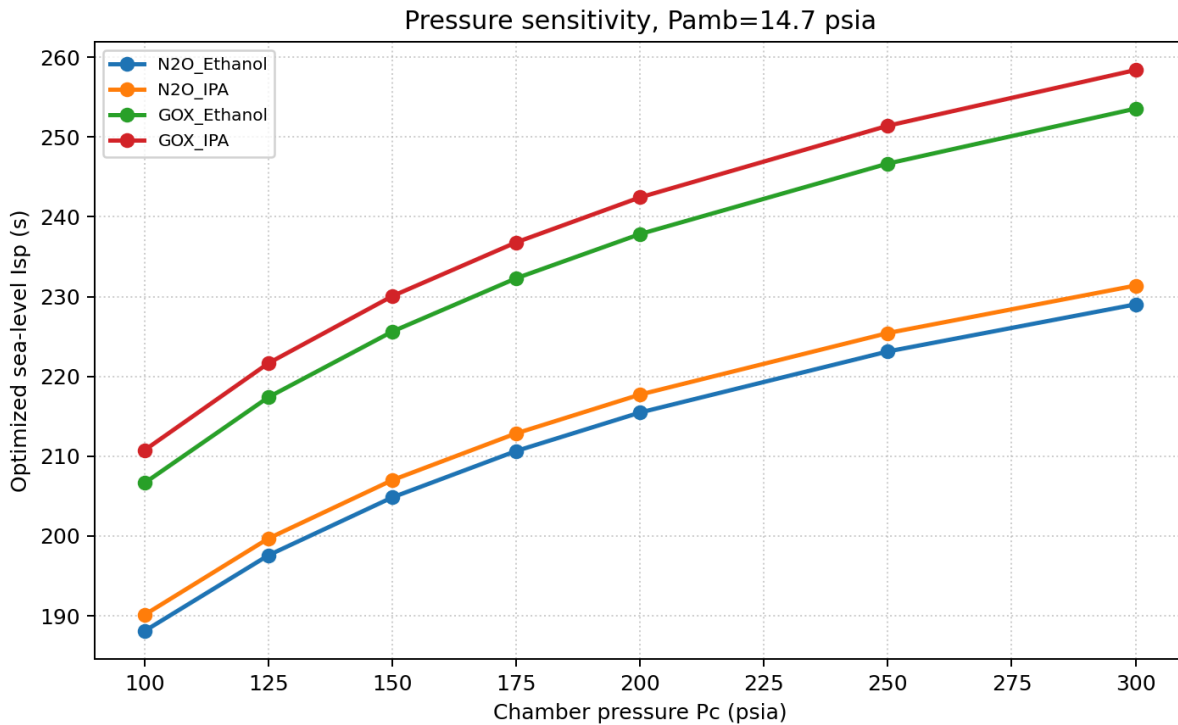


Figure 7: Sea-level ambient-corrected I_{sp} vs chamber pressure for the four alcohol survivors after screening (shifting equilibrium, $P_e \approx P_{amb}$ nozzle) [7].

2.7.1 Does higher P_c make GOX more reasonable?

A fair objection to the GOX storage penalty is that the thruster comparison was frozen at 150 psia, and GOX has higher ideal I_{sp} — so perhaps raising P_c shrinks the thruster and propellant mass enough to change the architecture trade. I checked that claim against the same pressure sweep rather than arguing it away.

On pure CEA performance, higher P_c does **not** close the GOX-to-N₂O gap. At sea-level-matched expansion, GOX/IPA minus N₂O/IPA is about 23 s at 150 psia, 25 s at 200 psia, and 27 s at 300 psia; GOX/ethanol minus N₂O/IPA is about 19 s, 20 s, and 22 s over the same pressures [7]. Absolute I_{sp} rises for every pair, but the **ordering is unchanged** and the GOX advantage over N₂O widens slightly rather than shrinks. Peroxide pairs were only computed at 150 psia; they were removed for sourcing and handling, not because a pressure sweep would have saved them.

On feed and storage physics, higher P_c makes regulated GOX **worse**, not better, for a pressure-fed first article. Injector inlet pressure still needs $\approx 1.2 P_c$ under the Huzel stiffness allowance [5], so a 300 psia chamber asks for

≈ 360 psia at the injector before valve and line losses. That raises the regulator delivery setpoint and therefore the usable blowdown floor on a fixed 2265 psia service cylinder: less of the fill is usable, so stored mass and cylinder class grow for the same burned oxygen. N_2O has its own high- P_c problem (cold-tank saturation pressure and valve authority), documented in Section 2.6.3, but that is a reason to keep P_c modest for a self-pressurized oxidizer — not a reason to switch to GOX. The conclusion feeds the selection below rather than assuming it: P_c steers thruster and tank **sizing**, not the propellant architecture decision. Higher chamber pressure does not reverse the GOX storage/regulator penalty, so it should not be used as a reason to pick GOX over a liquid oxidizer at this vehicle class.

2.8 Nozzle Behavior at Throttle

Every comparison above uses a nozzle matched at full thrust, but the hover mission lives at throttled setpoints. At fixed geometry and fixed mixture ratio, chamber pressure falls roughly with mass flow and exit pressure falls with it, $P_e/P_{\text{amb}} \approx \dot{m}/\dot{m}_{\text{full}}$ for a sea-level-matched design, so every throttled setpoint runs the nozzle overexpanded.

For the $\text{N}_2\text{O}/\text{IPA}$ baseline the penalty is modest but permanent: about 199 s at an 80% thrust setpoint (-8 s from the matched 207.0 s) and about 188 s near the $\approx 62\%$ landing throttle implied by the 1.2 liftoff margin and a $\approx 25\%$ propellant fraction (-19 s). Across the hover band from $\approx 83\%$ down to $\approx 62\%$ thrust the delivered band is roughly 189–201 s — a nearly flat 5–7% flight penalty that stacks on top of the delivered- I_{sp} knockdown carried in Section 2.4.

Flow separation is not close at those depths. Summerfield-class criteria put separation onset near $P_e \approx 0.35\text{--}0.40 P_{\text{amb}}$ [5], which this nozzle does not cross until roughly 25–30% thrust. A Snark-class 20% throttle floor [4] would sit below that line and would need an explicit separation check or a truncated nozzle; the throttle-demonstration requirement in Table 1 does not. None of this moves the propellant choice — every candidate carries the same overexpansion physics — but the flight-performance budget must use throttled I_{sp} , not the matched-point number.

2.9 Cryogenic Reconsideration: LOX

The no-cryogenics rule was written from fear of complexity, so before selecting I am testing it against numbers instead of instinct — especially since both surviving oxidizers carry a structural penalty: GOX pays in tank tare, regulator hardware, and storage volume; N_2O pays in two-phase calibration and thermal sensitivity. LOX removes both at once. It is a dense, single-phase liquid at the injector (about $1,141 \text{ kg/m}^3$ at its normal boiling point [38]), it stores at low dewar pressure instead of as 2265 psia gas, and it is the cheapest oxidizer I can document: FAR sells 230 L, roughly 262 kg, for \$540 — about \$2/kg — and a trade source reports \$200-class quotes for a 180 L dewar [8], [11].

It also has the deepest small-VTVL heritage of any candidate in this notebook, which the hard screen ignored. Masten's Xombie flew a regeneratively cooled LOX/IPA engine 227 times, Xodiac flew pressure-fed LOX/IPA, and Armadillo's Pixel and Mod vehicles flew pressure-fed LOX/ethanol through the Lunar Lander Challenge [39], [40], [41]. The propellant family with the most small hover-flight history is not $\text{N}_2\text{O}/\text{alcohol}$; it is LOX/alcohol.

The bill on the other side is operational, and it is real: cryo-rated valves and seals, thermal contraction and icing, a chilldown procedure, boiloff that forbids long holds and forces load-late operations, dewar logistics (deposit, rental, transport), the same CGA G-4.1 oxidizer cleanliness as GOX [26], and oxygen-enrichment PPE. Decisive at this scale: LOX is not usefully self-pressurizing, so a pressure-fed LOX system reintroduces the high-pressure pressurant bottle and regulator that the N_2O architecture avoids. It buys back GOX-style feed hardware and adds cryogenic operations on top of it.

The screen therefore stands for the first article, but as a recorded decision with a reopen condition rather than a closed door: if the N₂O calibration campaign runs past roughly 15–20 hot fires without converging, or the field thermal envelope cannot be held, LOX/IPA is the rational pivot with the most direct heritage, and the no-cryogenics requirement would be formally waived at that point.

2.10 Flight Mass Closure First Cut

The 1.2 liftoff margin turns each thrust bracket into a wet-mass cap: 17.0 kg at 200 N and 34.0 kg at 400 N. Subtracting the mapped N₂O storage from Section 2.4 (loaded bottle, tare plus 1.25× propellant load) and the fuel load leaves the budget for everything else — engine, structure, fuel tank and pressurant, landing gear, avionics — which I estimate at 6–9 kg for this vehicle class (project estimate, not a weighed bill of materials):

Case	Loaded N ₂ O bottle	Fuel	Left of cap	Verdict vs 6–9 kg
200 N, 30 s	9.4 kg (10 lb)	0.5 kg	7.1 kg	Marginal
200 N, 60 s	14.2 kg (15 lb)	1.0 kg	1.8 kg	Does not close
400 N, 30 s	14.2 kg (15 lb)	1.0 kg	18.8 kg	Closes with margin
400 N, 60 s	28.5 kg (2 × 15 lb)	1.9 kg	3.6 kg	Does not close

The conclusion is that the 60 s stretch burn is a **ground-demonstration** load, not a flight load: no corner of the envelope lifts a full 60 s propellant load and a realistic dry mass at $T/W = 1.2$. The flight article is either a 200 N vehicle flying ≈30 s partial loads with thin margin, or a 400 N-class vehicle — the same convergence AEL reached at 300 N for the same mission [4]. The delivered- I_{sp} knockdown in Section 2.4 takes a further ≈0.4–1.7 kg from these residuals, and GOX fails this check far harder: its mapped 26–49 kg steel cylinders exceed the wet-mass cap on their own. This is exactly the “final flight thrust converges in the engine-sizing section” outcome anticipated in Table 2, and it points toward the upper half of the thrust bracket.

2.11 Select Best Solution

The high GOX performance and simple single-phase metering are real, and the campaign figure in Section 2.4 shows GOX/IPA is genuinely the cheapest pure static-fire program, paying back its hardware premium in about 15 hot fires. Raising chamber pressure does not reverse that architecture trade: the pressure sweep keeps the same rank order and slightly widens the GOX-to-N₂O I_{sp} gap, while a higher regulated delivery pressure hurts GOX blowdown usable fraction (Figure 7). But the requirement is a hopper, and the quantified GOX penalty is really two distinct penalties. The budget penalty is \$1,190–\$1,870 of oxidizer-side hardware, nearly all of it regulator and oxygen-clean components rather than the cylinder. The flight penalty is the decisive one: mapped across the whole thrust/burn envelope, GOX storage carries about 7 kg of steel per kilogram of usable oxygen against about 1.5 kg/kg for a nitrous bottle, and a flight-weight composite oxygen vessel would fix the tare while making the budget worse. The high peroxide density is also real, but sourcing and decomposition risk dominate.

N₂O wins as an architecture, not as a propellant. Per kilogram it is the most expensive oxidizer in the finalist set, and Section 2.6.3 records that it is the only finalist whose injector and throttle cannot be closed analytically, so the selection explicitly buys a calibration campaign. What it buys back is that the storage bottle is also the pressurization system, the first fire is the cheapest to reach, and the closest public analog to this exact mission flew on it [2].

The judgment above is recorded as a weighted matrix so a reviewer can attack the weights instead of the prose. Scores are 1–5 and are generated with the same pipeline as the sizing and cost tables [19]. Ignition and start risk is scored as its own criterion rather than folded into development risk, because Section 2.6.3 shows it is the highest-energy failure mode in the set, not just a schedule item. N₂O/IPA wins while scoring **worst in class** on feed predictability, development risk, and ignition — the storage, cost, and heritage weights carry it — and the two ethanol alternates tie for second.

Criterion	Wt.	N ₂ O/IPA	N ₂ O/EtOH	GOX/EtOH	GOX/IPA	85% H ₂ O ₂ /IPA
Safety and access	0.15	4.0	3.5	3.0	3.0	1.5
Consumable and hardware cost	0.15	4.5	3.5	2.0	2.5	1.5
Storage mass and volume	0.15	4.0	4.0	1.5	1.5	5.0
Feed control and predictability	0.15	2.5	2.5	4.5	4.5	2.5
Development risk	0.10	2.5	2.5	4.5	4.5	2.0
Ignition and start risk	0.10	2.5	2.5	4.5	4.0	2.0
Direct heritage	0.10	5.0	4.0	3.0	2.0	1.0
Performance	0.05	3.0	3.0	4.5	5.0	3.0
Cooling / material margin	0.05	3.5	4.5	4.5	3.5	2.0
Weighted total	1.00	3.58	3.30	3.30	3.20	2.32

Table 3: Weighted selection matrix for the finalist set. Scores 1–5; weights and scores generated with the sizing/cost pipeline [19].

Decision Baseline propellant selection: N₂O / IPA. N₂O/IPA is not selected because it is analytically easy; it is selected because it best fits the full hopper constraint set: compact self-pressurizing oxidizer storage, cheap and accessible fuel, and the most relevant small-VTVL flight precedent. The design must explicitly budget the N₂O two-phase modeling and hot-fire calibration matrix as its own line item — about \$670–\$1,200 of propellant for a ten-fire matrix at 200 N, 60 s.

The best alternatives are conditional. **GOX/ethanol** is the cleanest technical alternate if avoiding two-phase N₂O behavior is worth the tank/regulator mass and cost. **N₂O/ethanol** remains the fuel backup if the project chooses to align with N₂O/ethanol literature or needs ethanol’s higher thermal conductivity for cooling margin. **GOX/IPA** is credible and has the best computed I_{sp} , but it is not the reference GOX fuel because the fuel choice does not solve the GOX storage/regulator problem and GOX/ethanol has stronger nearby ignition-validation evidence. **LOX/IPA** is the documented growth path: Section 2.9 records the reopen condition and the flight heritage that supports it.

3 Injector Design

3.1 Define the Problem

Design the injector for the selected baseline propellant pair N₂O/IPA at the Section 1.3 envelope. The injector must meter oxidizer and fuel into the chamber at the design O/F, atomize and mix them well enough for useful c^* efficiency, support the throttle-demonstration requirement, and be fabricable in aluminum or stainless under the ≤\$5,000 prototype-hardware budget.

The decision is not “maximize theoretical mixing only.” It is “pick an archetype I can machine, clean for N₂O service, and calibrate with water-flow and N₂O cold-flow,” accepting that flashing N₂O prevents a closed-form throttle map (Section 2.6.3, [4]).

3.2 Design-Point Inputs

These numbers are frozen from the propellant CEA optimum and the sizing envelope. Chamber diameter, L^* , and nozzle contour are **not** fixed yet; face packing that needs D_c is labeled provisional.

Quantity	200 N ref.	400 N scale	Source / note
Chamber pressure P_c	150 psia	150 psia	Sizing baseline Section 1.3.
O/F (mass)	5.10	5.10	N ₂ O/IPA CEA optimum [7].
$I_{sp,SL}$ (ideal CEA)	207.0 s	207.0 s	Ambient-corrected, $P_e \approx P_{amb}$ [7].
$\dot{m}_{tot}$	0.0985 kg/s	0.197 kg/s	From $\frac{T}{I_{sp,90}}$; scales with thrust at fixed I_{sp} .

$\dot{m}_{ox} / \dot{m}_f$	0.0824 / 0.0162 kg/s	0.165 / 0.0323 kg/s	O/F = 5.10; oxidizer is 83.6% of mass flow.
Throat diameter D_t	13.89 mm	19.6 mm	CEA geometry at matched nozzle [7].
IPA density (planning)	786 kg/m ³	same	≈20°C anhydrous IPA [30].
N ₂ O liquid density (planning)	750 kg/m ³ warm / 900 cool	same	Warm-tank planning vs cooler saturated liquid [18]; hot tanks hold less.
Ox/fuel volume ratio (warm)	≈5.35	≈5.35	Drives unlike-doublet momentum imbalance; see archetype trade.
Film-cooling reservation	0%	0%	Core sized at O/F 5.10 with no film allocation; a 10–20% fuel film would raise core O/F to ≈5.7–6.4 and force a fuel-orifice resize with the Cooling section.

Still open (not injector decisions)

Chamber diameter D_c , characteristic length L^* , contraction ratio, and nozzle contour are deferred. Where element layout needs a face diameter I use only a provisional band $A_c/A_t \approx 4\text{--}8$ (common pressure-fed practice band; final value belongs with chamber sizing) [5], [42]. Injector **orifice areas** do not require D_c ; **pitch and ring layout** do.

3.3 Injector Requirements

Item	Class	Requirement / design rule
Fuel metering	Hard	Size IPA with single-phase incompressible (SPI) orifice equation; freeze C_d from water-flow, not handbook alone.
Oxidizer metering	Hard	Do not size N ₂ O with SPI alone. Bracket SPI (upper flow / lower area), approximate HEM (lower flow / higher area), and Dyer/NHNE design estimate; require empirical cold-flow map [33], [34], [35] Section 2.6.3.
Injector stiffness (fuel)	Design rule	Target fuel injector $\Delta P/P_c \approx 0.15\text{--}0.20$ at full thrust as a classical isolation/stiffness band [5].
Injector ΔP (N ₂ O)	Design rule	Hold a planning ΔP band, but do not assume $\dot{m} \propto \sqrt{\Delta P}$ after flashing; Snark found N ₂ O injector drop nearly independent of massflow [4].
Materials	Hard	Aluminum alloy or stainless steel primary wetted structure Table 1.
Cleanliness	Hard	Oxidizer-wetted hardware free of hydrocarbon contamination; N ₂ O ullage decomposition risk if oil/grease present [27].
Pressure safety	Hard	Custom manifolds/face: burst FoS ≥ 4 vs MEOP Table 1.
Fabrication	Hard	First article machinable with common tools; avoid processes that force the whole injector into specialty diffusion-bond/platelet shops.
Throttle	Test req.	Support full and reduced thrust setpoints; primary throttle authority is the run valve, not a movable injector element, unless a later pintle growth path is chosen [4].
Interfaces	Layout	Separate ox/fuel manifolds with no interpropellant leak path; provisions for ignition and P_c tap; do not block later TVC envelope Table 1.

3.4 Brainstorm: Injector Archetypes

Taxonomy below follows standard liquid bipropellant injector families in Huzel and Huang and NASA SP-8089 (liquid rocket engine injectors monograph) [5], [42]. Pros/cons are scored for **this** N₂O/IPA pressure-fed hopper, not for a general LOX/RP booster.

Archetype	Mechanism	Why it can work here	Why it may fail here
Showerhead (non-impinging)	Parallel axial jets; mixing by diffusion/shear.	Simplest to drill; common hybrid N ₂ O practice for ox-only.	Biprop mixing is weak without impingement; long L^* or film risk [42].
Unlike-doublet	Fuel jet hits oxidizer jet; sheet then atomizes.	Best initial mixing of the impinging family when stream momenta match [42].	O/F mass 5.1 and warm volume ratio ≈5.35 → severe momentum imbalance; reactive-stream separation risk [43].

Like-doublet (self-impinging)	F-F and O-O pairs form fans; fans mix downstream.	Each propellant atomizes on itself; tolerates unequal O/F; good stability reputation vs unlike [42], [43].	Needs enough face area and secondary mixing length; more holes than a pintle.
Triplet (e.g. F-O-F)	Three jets meet; often fuel-rich outer.	Can improve mixing vs doublet; used on some storables [42].	Three-way drill angles harder; still sensitive to ox jet quality when N ₂ O flashes.
Pentad / multi-orifice element	Central jet + surrounding ring of jets.	High mixing potential at large scale [42].	Pattern density and drill complexity high for a 200 N first article.
Splash-plate / cup	Jets hit a surface, form sheet.	Can work with coarse orifices.	Face heat and erosion; less common for reusable small hoppers [42].
Pintle (fixed)	Central radial sheet impinges annular axial sheet.	Deep throttle heritage (LMDE class); single element; good for unequal flows [43], [44].	Concentric machining and skip-distance control; N ₂ O in annulus or slots still flashes [34].
Pintle (movable / throttling)	Variable annular gap for deep throttle.	Best throttle authority in the injector itself [44].	Moving ox/fuel seal in a hot, N ₂ O-clean environment is not a first-article risk I will buy.
Coaxial shear	Annular + core jets, no swirl.	Simple axisymmetric machine work.	Liquid-liquid shear mixing weaker than impingement at low ΔP [42].
Coaxial swirl	Tangential entries form hollow cones.	Fine atomization possible.	Swirler passages clog-sensitive; two-phase N ₂ O swirl poorly predicted [42].
Platelet / etched multilayer	Photo-etched bonded stacks; arbitrary passages.	Excellent feature control at aerospace budgets.	Diffusion-bond/platelet process cost and lead time conflict with $\leq \$5k$ first article Table 1.
Hybrid face (e.g. ox showerhead + fuel impinging)	Combine families on one face.	Can tailor ox vs fuel.	Two design problems and two calibration maps; complexity without proven need.
Gas-liquid coaxial (GOX note)	Gaseous ox core/annulus with liquid fuel.	Relevant only if GOX alternate is later chosen [37].	Not the N ₂ O/IPA baseline; listed for completeness.

3.5 Hard Screens

Screen result: remove Platelet / etched multilayer and any design that **requires** diffusion bonding or ultra-fine EDM as the only viable build path — process cost and lead time violate the first-article budget and material-access intent Table 1 [42].

Screen result: remove Movable throttling pintle as baseline — deep throttle heritage is real [44], but a moving oxidizer seal, actuation, and calibration stack on top of already non-analytic N₂O flow is the wrong first risk [4] Section 2.6.3.

Screen result: remove Pure showerhead biprop as the sole mixer — SP-8089-class guidance treats non-impinging patterns as weaker mixers; I will not bet the first c^* campaign on diffusion-only mixing at this L^* uncertainty [42].

Screen result: remove Coaxial swirl and splash-plate as baseline — swirl passages plus two-phase N₂O are a prediction dead-end for a first map; splash plates add face heat/erosion without a compelling small-hopper precedent I can cite [34], [42].

Screen result: remove Pentad / high-order multi-orifice elements — unnecessary element complexity at 200 N when doublets or a fixed pintle already cover the trade space [42].

Survivors for quantitative trade: **unlike-doublet**, **like-doublet**, **fixed pintle**, and **triplet (F-O-F)** as a complexity check. Hybrid faces and gas-liquid coaxial remain contingency notes only (GOX path), not N₂O baseline candidates.

3.6 Quantitative Flow Models

Shared orifice math used for every survivor. Units SI in calculation; tables in engineering units.

Fuel (IPA) – SPI. Single-phase incompressible orifice flow [5]:

$$\dot{m}_f = C_{d,f} A_f \sqrt{2\rho_f \Delta P_f}$$

Planning $C_{d,f} = 0.70$ until water-flow; $\rho_f = 786 \text{ kg/m}^3$; target $\Delta P_f = 0.20P_c = 30 \text{ psi}$ at 200 N full thrust (top of the 15–20% band) [5].

Oxidizer (N₂O) – SPI / HEM / Dyer bracket. SPI is only an upper mass-flow bound for a fixed area when the liquid can flash [33], [34]. Approximate HEM here is a **planning lower bound**: $\dot{m}_{\text{HEM}} \approx 0.35 \dot{m}_{\text{SPI}}$ at the same A and ΔP , an engineering factor in the band where short-orifice saturated N₂O data sit between SPI and deep equilibrium limits in Waxman-class work – **not** a first-principles HEM integration [34]. Dyer/NHNE design estimate [33], [35]:

$$\dot{m}_{\text{Dyer}} = \frac{\dot{m}_{\text{SPI}} + k \dot{m}_{\text{HEM}}}{1 + k}$$

with planning $k = 1$ (saturated, short-orifice mid blend). Required **area** for a target $\dot{m}$ is therefore largest under HEM, intermediate under Dyer, and smallest under SPI (SPI under-sizes the hole if flashing reduces discharge).

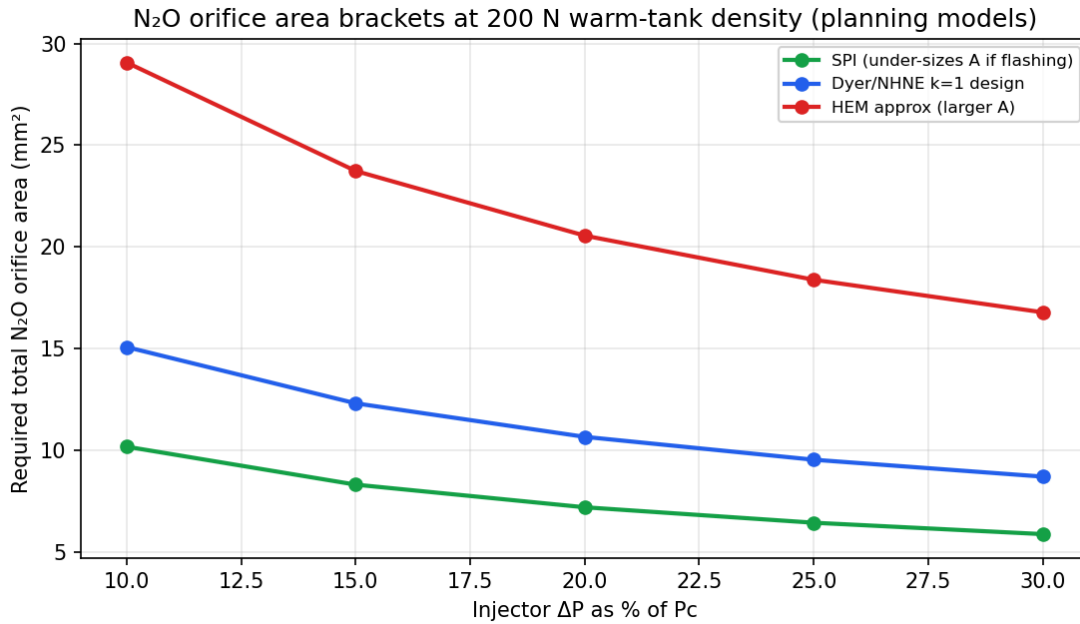


Figure 8: Required total N₂O orifice area vs injector $\Delta P/P_c$ at 200 N, warm-tank density 750 kg/m^3 , $C_d = 0.65$. SPI under-sizes area if flashing; HEM approx over-sizes; Dyer $k = 1$ is the planning design curve [7], [34].

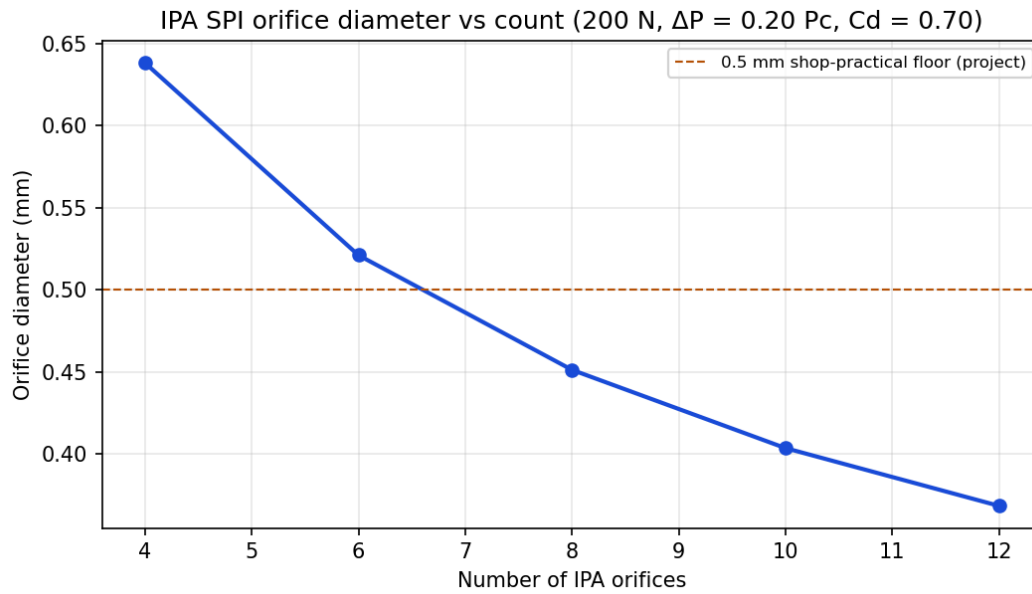


Figure 9: IPA SPI orifice diameter vs hole count at 200 N, $\Delta P = 0.20 P_c$, $C_d = 0.70$. Project shop floor marks 0.5 mm as a clogging/practical limit (assumption, not a standard).

At the 200 N, $\Delta P = 0.20 P_c$ planning point the generated tables give:

Stream	$\dot{m}$	Model	Required total A
IPA	0.0162 kg/s	SPI	1.28 mm ²
N ₂ O	0.0824 kg/s	SPI (min A if no flash)	7.2 mm ²
N ₂ O	0.0824 kg/s	Dyer $k = 1$ design	10.7 mm ²
N ₂ O	0.0824 kg/s	HEM approx (max A)	20.6 mm ²

Exact CSV rows live in rocket_outputs/data/injector_orifice_sizing_*.csv [45]. 400 N at the same P_c and ΔP fraction doubles areas (diameters scale $\sqrt{2}$).

Impinging geometry parameters (when an impinging survivor wins). NASA SP-8089 and subsequent practice commonly use 60° included impingement angle for like- and unlike-doublets [42], [46]. L/d of a few orifice diameters is a usual short-tube starting point; I adopt a planning $L/d \approx 3$ pending water-flow C_d [5]. Unlike-doublet design also wants comparable jet momenta; with volume ratio ≈ 5.35 that condition is badly violated unless the pattern is heavily ox-sided — a structural reason unlike-doublet scores poorly below [42].

3.7 Analyze Survivors

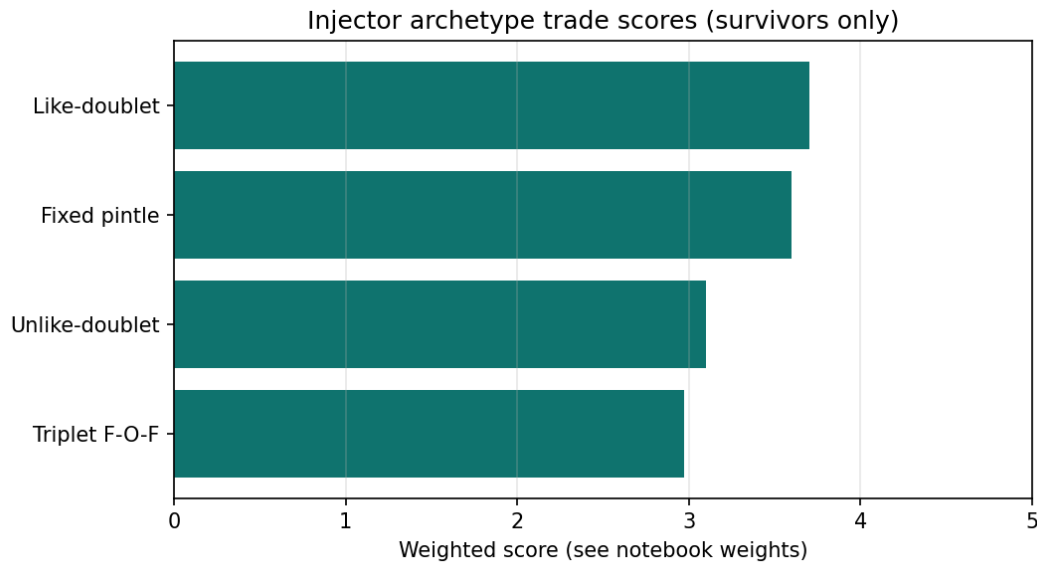


Figure 10: Weighted scores for the four hard-screen survivors. Weights: N_2O two-phase tolerance 0.25, fab simplicity 0.20, mixing 0.15, unequal-O/F suitability 0.15, throttle fit 0.15, stability heritage 0.10 [45].

Weights intentionally favor N_2O honesty and shop reality over pure mixing bragging rights — matching the propellant section’s risk posture.

3.7.1 Unlike-doublet

Unlike-doublets give strong **initial** mixing when fuel and oxidizer jet momenta are comparable [42]. Here they are not: warm-tank volume ratio $ox/fuel \approx 5.35$ at $O/F = 5.10$. Reviews note asymmetric sprays and reactive-stream separation risk when properties and momenta differ strongly [43]. Flashing N_2O makes the oxidizer jet even less like a steady liquid spear [34]. Fabrication of angled pairs is familiar, but I would be calibrating a pattern that is already on the wrong side of the momentum-ratio rule. **Trade score: lowest of the four survivors.**

3.7.2 Triplet (F–O–F)

Triplets can improve mixing relative to a single doublet and appear in storable-engine practice [42]. They inherit the same unequal-jet and N_2O flash problems as unlike elements, with harder three-axis drill geometry. Not worth the complexity on a 200 N first face. **Eliminated as baseline; not carried as backup.**

3.7.3 Fixed pintle

Fixed pintles impinge a radial sheet with an annular sheet and have deep throttle **heritage** (LM descent engine class; later LOX/RP engines) [43], [44]. They handle unequal propellant flows more gracefully than unlike-doublets and put throttle authority in geometry if the gap is varied — but I already screened **movable** pintles. A **fixed** pintle still helps throttle by keeping a stable spray structure while the **valve** meters, which matches Snark’s valve-command-linear P_c observation [4]. Downsides: concentricity, skip distance, and putting flashing N_2O through an annulus or slots is still two-phase orifice physics [34]. Machining is lathe-centered rather than multi-hole drill jigs. **Strong alternate, especially if early hot-fires show face-pattern mixing or heat problems.**

3.7.4 Like-doublet

Like-doublets atomize each propellant against itself (F-F and O-O), then mix the fans in the chamber. Literature treats them as more forgiving for combustion stability and for avoiding reactive impingement issues than unlike-doublets [42], [43]. Critically, they **do not require** matched F and O jet momenta: I size the ox pairs for the Dyer N₂O area and the fuel pairs for SPI IPA area separately. Drill count is higher than a pintle but each hole is a straight (angled) gun-drill problem I can jig. Throttle remains valve-side, which is what the N₂O evidence supports [4]. **Highest weighted score in Figure 10.**

3.8 Select Best Solution

Decision Baseline injector archetype: like-doublet impinging (self-impinging F-F and O-O pairs). Chosen because it separates atomization from interpropellant momentum matching, fits O/F = 5.1 volume asymmetry, stays within Al/SS drill-and-jig fabrication, and keeps throttle authority in the run valve where Snark already showed nearly linear P_c command [4], [42], [43].

Ranked backups. (1) **Fixed pintle** if like-doublet face heat, packing, or c^* efficiency disappoint after cold-flow/hot-fire — throttle-structure heritage is the reopen reason [44]. (2) **Unlike-doublet** only if cold-flow shows I can force acceptable ox/fuel momentum by extreme area splitting **and** flash-boiling does not destroy the sheet — unlikely, kept as literature comparison only.

GOX alternate note: if the program ever pivots to GOX/alcohol, gas-liquid coaxial or unlike elements become more attractive because the oxidizer is single-phase gas [37]; that is a different injector, not a bolt-in of the N₂O face.

3.9 Detailed Specifications of the Chosen Archetype

All dimensions below are **preliminary design values** for the 200 N, $P_c = 150$ psia reference. They come from SPI fuel + Dyer ($k = 1$) N₂O area budgets at $\Delta P = 0.20P_c$ and must be revised after water-flow C_d and N₂O cold-flow [45].

Item	Preliminary value	Basis
Pattern	Like-doublet; separate F-F and O-O pairs	Selection above [42].
Included impingement angle	60°	Common doublet practice [42], [46].
Fuel pairs / orifices	3 pairs / 6 holes	SPI area 1.28 mm ² total $\rightarrow d_f \approx 0.52$ mm each; three fuel pairs interleave 1:2 with the six ox pairs for fan overlap [45].
Ox pairs / orifices	6 pairs / 12 holes	Dyer area 10.7 mm ² total $\rightarrow d_o \approx 1.06$ mm each [45].
Orifice L/d (both)	≈ 3	Short-tube planning; set final from measured C_d [5].
Planning C_d	Fuel 0.70; ox 0.65	Until water-flow / cold-flow; ox lower pending flash.
Target ΔP at full thrust	30 psi (0.20 P_c) each side as manifold-to-chamber planning drop	Fuel stiffness band [5]; N ₂ O drop may not track $\dot{m}^2$ [4].
Face layout	Alternating rings or sectors of O-O and F-F pairs; pitch $\geq 3d$ local (project packing assumption until D_c fixed)	Provisional with A_c/A_t band; finalize with chamber diameter.
Manifolds	Separate ox and fuel ring manifolds behind the face; no shared cavity	Interpropellant leak prevention [42].
Inlet ports	AN/JIC or equivalent on manifold block	Hard connector rule Table 1.
Materials	SS316 or Al 6061-T6 face/manifolds per compatibility and weld/braze plan	Hard material rule; final pick with structural section.

Seals	Oxidizer-compatible elastomers or metal seals; no hydrocarbon grease on ox side	[27].
Ignition provision	Boss or film-cooled torch/spark port on chamber, not through a propellant orifice	Keep injector holes for propellants only.
Instrumentation	P_c tap on chamber; optional manifold P_{ox} , P_f taps	Required for c^* and ΔP data.
Throttle concept	Fixed orifices; primary throttle = run valves; build empirical map vs tank T , P	[4] Section 2.6.3.
400 N scale	Areas $\times 2$ at same P_c and ΔP fraction; or raise P_c later	Thrust scale at fixed I_{sp}

▷ N₂O sizing honesty

If cold-flow shows discharge closer to HEM than Dyer, ox holes must grow (or ΔP rise). If closer to SPI, holes can shrink. Do not laser-cut the face until at least one N₂O cold-flow campaign on a prototype orifice plate has bounded C_d effective [34], [35].

Shop notes (project assumptions, not standards). Prefer gun-drilled or reamed holes over punched sheet for C_d repeatability; deburr exits; keep entrance edge consistent across the set; drill-angle fixture tolerance goal $\pm 1^\circ$ on impingement half-angle until data say otherwise. Minimum finished fuel diameter stays above 0.5 mm to limit clogging – the 0.52 mm planning size sits just above that floor; if water-flow pushes fuel holes smaller, drop to 2 pairs (0.64 mm) rather than go under it.

Open items before freezing hardware. (1) Water-flow C_d vs $\mathfrak{R}$ for fuel and for water-as-proxy ox geometry. (2) N₂O cold-flow $\dot{m}(T_{\text{tank}}, P_{\text{tank}}, \text{valve})$ on a spare face. (3) Chamber D_c so pitch/rings lock. (4) Film-coolant ring decision with Cooling section – any film fraction comes out of the 0.0162 kg/s fuel stream and moves core O/F, so fuel orifices resize with it. (5) Structural FEA of face under MEOP.

3.10 Verification and Calibration Plan

- Water-flow (fuel circuit, and ox geometry with water).** Measure C_d for each orifice class; compare to 0.70/0.65 planning values; reject faces with $>10\%$ element-to-element scatter (project threshold).
- N₂O cold-flow.** Map mass flow vs tank temperature/pressure and valve command; overlay SPI / Dyer / HEM predictions; update orifice diameters if outside $\pm 15\%$ of target $\dot{m}$ at the design tank state (project threshold).
- Hot-fire matrix.** Reuse the propellant-section budget (ten 200 N, 60 s-class fires): record c^* efficiency vs ideal 1590 m/s, O/F from tank weigh-back, chug/buzz notes, and face condition [4], [7].
- Pass criteria (first article).** Repeatable ignition; no hard start that damages hardware; c^* efficiency high enough for hover math with the 1.2 liftoff margin after real I_{sp} ; no face burn-through; throttle steps between full and reduced setpoints without divergence Table 1.

Until those data exist, the like-doublet geometry above is a **design baseline**, not a flight-certified drawing.

4 Ignition System Design

Scope recorded now; design deferred. The baseline concept is a spark-torch-class igniter with its own propellant metering and calibration, an ignition-proven-before-main-oxidizer-valve interlock, and a bounded start transient. Hard-start avoidance is the design driver, because the N₂O ullage is monopropellant-capable and a slow or failed light feeds it (Section 2.6.3). The chamber ignition boss reserved in the injector specification is the interface this section will design to. Igniter propellant choice, sizing, and the start sequence are not yet developed.

5 Feed System Design

6 Cooling Analysis

7 Structural Analysis

8 CAD Design

9 Simulation

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